Advanced Natural Language Processing (968G5): Practical Assignment (Max 5000 words) - Propaganda Detection

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Word count: XXXX

Introduction

Propaganda constitutes a biased representation of information which is intended to persuade others of a particular belief or worldview (Cambridge University Press & Assessment, n.d.). Whilst not necessarily false, propaganda risks engendering beliefs in individuals which do not adhere to matters of fact and that may serve to drive support for policies and societal attitudes which cause harm to indiviudals and groups, such as by reducing intent to vaccinate (Loomba et al., 2021). Social media provides a strong vector for propaganda to spread. As an example, this is particularly consequential when taking the form of state-sponsored "disinformation campaigns" (DiResta & Goldstein, 2025). Flagging propagandistic content to social media users may help to combat propagandistic content by encouraging users to be more critical of such content and reduce belief adoption based on misleading propaganda (Martel & Rand, 2023; Mena, 2026; but also see Grady et al., 2021). A reliance upon human-judges requires a set of impartial assessers to review (ideally) every social media post and flag its propagandistic nature. Such an approach is approximated by X in the form of community notes ("Community Notes," 2026), but whilst efficacy is generally good (Allen et al., 2024) coverage is far from ideal (Maldita.es, 2025), particularly during important political events (Center for Countering Digital Hate, 2024). Instead, automated detection may help to improve propaganda flagging.

This project aims to address two tasks in propaganda detection. Firstly, two approaches to classifying known propagandistic texts as one of 8 propaganda variants will be evaluated, namely, a multinomial naive Bayes (MNB) model using a unigram-bigram bag-of-words representation, and a Bidirectional Encoder Representations from Transformers BERT (Devlin et al., 2019) model, fine-tuned to this project's propaganda text dataset.

Real-world applications of propaganda technique text classification will likely not know in advance which texts contain propaganda, nor will the location of the propagandistic snippet be known. Thus, task 2 develops on the task 1's classification problem by also identifying where the snippet exists within each text. Again, two approaches will be compared. The first of these approaches will initially use a MNB to classify each text as `propaganda` or `not_propaganda`. Then, with the texts which are classified as containing propaganda, each possible n-gram from each text will be enumerated

Exploratory Data Analysis

In machine learning classification tasks, the distribution of classes in the training data can influence the manner in which models discriminate between classes. For instance, suppose a class in the training data equates to 95% of all labels present. A model which only predicts this single class will achieve an accuracy of 95%. Thus, it is appropriate to evaluate the class distribution before considering accuracy as an appropriate evaluation metric for text classification.

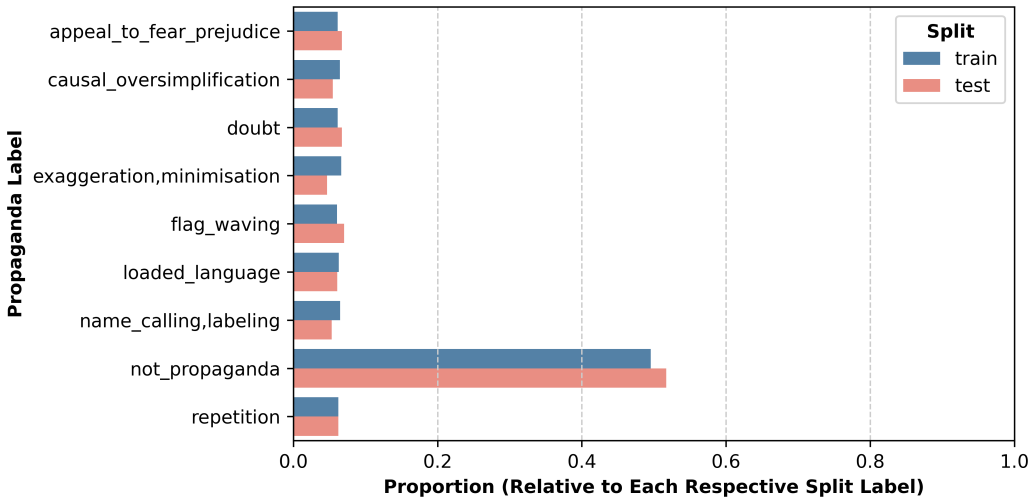


Figure 1. Comparison of Multiclass Distribution Between Train and Test Datasets

Figure 1 shows that whilst `not_propaganda` instances dominate the class distribution ($\approx$ 50%), the class distibution amongst propagandistic labels is fairly uniform. This would suggest that for task 1, where the texts are known to be propagandistic, the use of accuracy to evaluate overall model performance would be appropriate. However, for task 2, where the propagandistic nature of the text must be determined via span identification, it would be more appropriate to evaluate overall model performance using a weighted F1-score, which computes the F1-score (harmonic mean of precision and recall) for each class and then averages these scores weighted by the number of true instances (support) in each class.

As both task 1 and 2 involve propaganda sequence classification, Table 1 outlines the most commonly occurring unigrams and bigrams amongst snippet and context strings from texts in each propaganda class. This excludes English stop words, as defined by the `ntlk` library. Variation in n-gram occurrence across classes can provide some prior expectation around which kinds of n-grams may be most discriminiative in classification.

Label	Snippet	Context
appeal_to_fear_prejudice	church, one, would, the church, people, u, u s, could, us, god	said, many, letter, us, trump, plague, could, people, bishops, even
causal_oversimplification	one, god, cia, church, the cia, the church, trump, clearly, want, would	said, one, would, security, like, words, national, trump, national security, man
doubt	guardian, the guardian, people, even, evidence, know, trump, never, story, church	said, church, question, also, cannot, kavanaugh, people, viganò, pope, catholic
exaggeration,minimisation	world, one, history, ever, thing, the world, church, without, even, life	one, trump, said, president, people, one of, church, even, many, former
flag_waving	american, people, america, american people, the american, country, americans, protect, nation, u	trump, said, continue, president, public, put, mayors, iran, cia, safety
loaded_language	church, people, life, smear, whoa, scandal, pope, last, agenda, damn	trump, church, said, would, also, new, many, the church, one, people
name_calling,labeling	trump, leftist, left, america, disgrace, anti, infidels, partisan, priest, disgrace to	trump, one, would, church, news, assange, catholic, people, like, must
repetition	aliens, people, autonomy, trump, illegal, illegal aliens, muslim, invaders, never, become	one, said, parents, trump, many, obama, school, president, immigration, even

Table 1. Most Common Unigrams/Bigrams by Label

We can see from Table 1 that each class does appear to have some variation in commonly occurring n-grams. For instance, `flag_waving` appears to have multiple n-grams involving the word `america` in its snippets which is distinct from the other classes. Furthermore, we can see that n-grams relating to immigration appear under `repetition` more so than other classes, which may be indicative of how immigration-related propaganda tends to lean towards the use of repetition compared to other topics.

On the other hand, we can also see that certain n-grams are unlikely to provide much discriminative information, such as `trump` which appears across most classes, both snippets and context strings alike. This is potentially an artefact of the dataset being biased towards American political news articles, which likely involve mentioning the U.S. president.

Methods

Task 1

Approach A: Bag-of-Words with Multinomial Naive Bayes

Given a set of known propagandistic texts, the propagandistic snippet can be identified from the `<BOS>` and `<EOS>` tags. The remainder therefore constitutes the sentential context. For each of these substrings within each propagandistic text, we can construct a bag-of-words (BOW) representation. This involves constructing a dictionary where each key is a unique word and each value is the number of times that word occurs within the text. This idea can be extended to n-grams, whereby each key becomes a unique n-gram.

However, the distribution of words within any given text is not uniform. Some words appear more frequently across texts than others, and these more frequent words tend to be the same set of words across texts. This phenomenon is referred to as Ziph's law, whereby the product of frequency of a word and its rank amongst all other words remains approximately constant. As such, a BOW representation using raw frequencies will tend to have higher frequencies for the same set of words, regardless of the text in question. Given that this task involves assigning a propaganda label to a given text, it would be more useful to overweight infrequent n-grams as these are more likely to offer discriminative information during classification than frequently occurring n-grams which are relatively common to any given text.

To achieve this, the present BOW representation will use Term Frequency-Inverse Document Frequency (TF-IDF) to represent n-gram frequencies. TF-IDF simply weights each n-gram's observed frequency within a single text by the logarithmic inverse of its frequency across all texts within the sample. As such, a given text may include the word "the" 3 times, but given that "the" occurs commonly across all texts, this frequency is then downweighted. By contrast, a word which is rare across all texts, but is found within the given text will have its frequency more heavily weighted. This helps to differentiate BOW representations by overweighting their more discrimnative n-grams, which will be more informative in classification than using raw frequencies.

As such, a unigram and bigram BOW representation is used for both the snippets and the sentential context for each data instance. To incorporate sentential context, its BOW representation is concatenated to that of the snippet's to produce a single vector representation for each text (where the vector is just a vector of the text's TF-IDF frequencies for each n-gram). The incorporation of sentential context, rather than using snippets alone, may help to differentiate between snippets with similar BOW representations but fall under different class labels. As Table 1 shows, `church` appears commonly across `appeal_to_fear_prejudice` and `causal_oversimplification` snippets, whereas their common context n-grams show a greater variety of n-grams.

With this BOW representation, a multinomial naive Bayes (MNB) model can be fitted to a training set of propaganda label and BOW vector pairs. MNB is a supervised classification technique used to estimate the probability of some document d belonging to some class c.

$$P(c|d) \propto P(d|c)P(c) \tag{1}$$

We can represent a document d as a sequence of n words $d = w_1, \ldots, w_n$. To find the likelihood of observing document d given class c ($P(d|c)$), we can assume that the probability of each word in the sequence d, given class c, occurs indepdently of every other word in the sequence:

$$P(w_1, \ldots, w_n|c) = P(w_1|c) \times \cdots \times P(w_n|c) \tag{2}$$

The prior $P(c)$ refers to the probability of observing class c, which can be estimated from the class distribution in the training data:

$$P(c) = \frac{\text{freq}(d_c)}{d_{total}}$$

During classification, we predict the class of each document by assigning the most probable class label, given some document d with frequencies f of each ith n-gram in the training corpus vocabulary V.

$$\hat{y} = \arg\max_{c \in C} \left[\log P(c) + \sum_{i \in V} f_i \log \hat{P}(w_i|c) \right] \tag{3}$$

Given that class estimates require a summation over the entire training corpus vocabulary, some conditional probabilities could be 0 as certain n-grams may never appear in a given class. This would prevent class predictions as probability estimates for class membership would be 0 due to multiplying the conditional probabilities of each n-gram (or $-\infty$ in log space). To avoid this issue, Laplace smoothing can be used to increase the frequency of every conditional event by 1, ensuring every conditional probability is > 0.

$$\hat{P}(w_i|c) = \frac{\text{freq}(w_i, c) + 1}{\sum_{w \in V} (\text{freq}(w, c) + 1)} \tag{4}$$

Evaluation Metrics

As shown in Figure 1, whilst `not_propaganda` dominates the class distribution, propagandistic classes are distributed uniformly. As such, it is appropriate to use accuracy as an evaluation metric. Furthermore, to understand model performance within each class, a confusion matrix will be used to compare precision and recall and identify poorly performing classes.

To understand how this MNB model makes class distinctions, we can use the estimated log probabilities of each n-gram given each class to compare each feature's probability of belonging to class i with its average probability of belonging to every other class. Features which have a higher probability of class i than their average probability across the other classes suggest the feature is more unique to that given class and may thus be more discriminative for assigning class labels to texts. Table 3 shows the top 10 most discriminative features for each class, which can be contrasted against Table 1.

Approach B: Fine-tuned DistilBERT

Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2019) is a type of transformer architecture which uses an encoder to generate representations of text from large corpora, pre-trained on masked language modelling and next sentence predictions tasks, respectively. BERT uses three word embeddings to represent WordPiece tokens: token embeddings, segment embeddings and position embeddings. Each segment is separated by a [CLS] token and sentences within segments are separated by [SEP] tokens. BERT processes text bidirectionally to account for prior and future context and generates contextual embeddings of each token, with the [CLS] token representing a summarised representation of the entire segment. BERT can then be fine-tuned to novel corpora to perform additional NLP tasks such as sequence classification. Fine-tuning can be used to generate [CLS] embeddings for each segment of text, which can be passed to a feedforward neural network to classify that [CLS] token.

This project will use DistilBERT (Sanh et al., 2020), which is a smaller and more efficient variant of BERT, which only slightly underperforms BERT across standard benchmarks but has 40% fewer parameters. Unlike the MNB classifier, this approach will use the unmodified `tagged_in_context` training data, whilst adding the `<BOS>` and `<EOS>` as special tokens, akin to the [SEP] token.

Hyperparameters

Table 2 outlines the hyperparameters used for this approach.

Parameter	Value
batch_size	8
learning_rate	2e-05
max_token_length	512
num_epochs	3
pretrained_bert_model	distilbert-base-uncased
validation_test_size	0.2
weight_decay	0.01

Table 2. Task 1 DistilBERT Hyperparameters

Batch size is used to determine the number of data instances which pass through the model

before its weights are updated. Higher batch sizes many improve weight stability but require more memory, whereas smaller sizes sacrifice stability but are less intensive on memory. A batch size of 8 was used as this enabled faster training.

The learning rate is a weighting factor applied to the loss gradient. When too high, training losses diverge from the global minimum but too low and the model fails to update weights sufficiently to approach the global minimum. A typical learning rate of $2e-5$ was used as this is a typical value used for fine-tuning tasks with BERT (Devlin et al., 2019).

BERT model use a maximum sequence length of 512 tokens, which is retained in this project. A lower maximum sequence length would improve training speed, but loses information for longer sequences. Thus, keeping this hyperparameter at 512 ensures training examples contain maximal information.

The number of epochs was determined during training by evaluating the validation loss. Fine-tuning tasks typically do not require many epochs (i.e. < 5) to converge, but inspection of validation losses suggested that 3 was optimal as subsequent epochs resulted in increasing valdiation loss, indicative of overfitting. 20% of the training data instances were held out to evaluate validation loss.

Lastly, weight decay is used to implement L2 regularisation, which adds an additional penalty term to the model's loss function, equivalent to the sum of the squared model weights. The weight decay therefore acts as a weighting on this penality term. High weight decays risk underfitting, and so a value of 0.01 is used.

Loss Function

The `AutoModelForSequenceClassification` class from the Hugging Face `transformers` library uses unweighted cross entropy loss as its default loss function when the `num_labels` argument is passed with a value > 1. Cross entropy loss measures the expected negative log-likelihood of the true probability distribution $p(x)$, given a predicted distribution $q(x)$. Lower cross entropy loss means predictions made by a model with $q(x)$ are close to those produced under $p(x)$, indicating $q(x)$ aproximates $p(x)$.

$$H(p,q) = -\sum_{x \in X} p(x) \log q(x) \tag{5}$$

As each data instance can only have one class label, $\sum_{x \in X} p(x)$ becomes a one-hot encoded vector across each class, which therefore simplies Equation 5 to simply be the negative log of the model's predicted probability for the true class ($H(p,q) = -\log q(x)$).

Evaluation Metrics

As with Section , accuracy will be used to evaluate model performance holistically, whilst a confusion matrix will highlight class-level precision and recall underperformance. However, unlike MNB, evaluating discriminative features for each class is less feasible with BERT given than DistilBERT uses contextual embedding representations of WordPiece tokens,

rather than a bag-of-words representation. Thus, n-grams in MNB constitute independent model features used in predictions, whilst DistilBERT's token representations are directly dependent on surrounding tokens and thus cannot be extracted into atomised model features.

Task 2

Approach A: BIO Tagging with DistilBERT

With task 2, the texts are no longer assumed to be propagandistic and thus the potential propagandistic snippet must be idenfitied before enabling classification. An approach which helps combine these two tasks into a single fine-tuning BERT task is BIO tagging.

BIO tagging involves assigning labels to tokens within a sequence. A token tagged with `B` indicates that the token is the beginning of a span. `I` then indicates subsequent tokens which belong to that span, whilst `O` shows tokens which are outside the span. Applying this tagging to the training and testing data provides token-tag pairs which can be used for training token classification, which DistilBERT can be fine-tuned to perform.

To combine the sequence labelling, we can also append an entity type to each `B` and `I` tag, indicating what kind of span it is labelling. These entities can correspond to each of the 8 propaganda classes in the dataset, with tokens from `not_propaganda` spans simply being labelled with `O`.

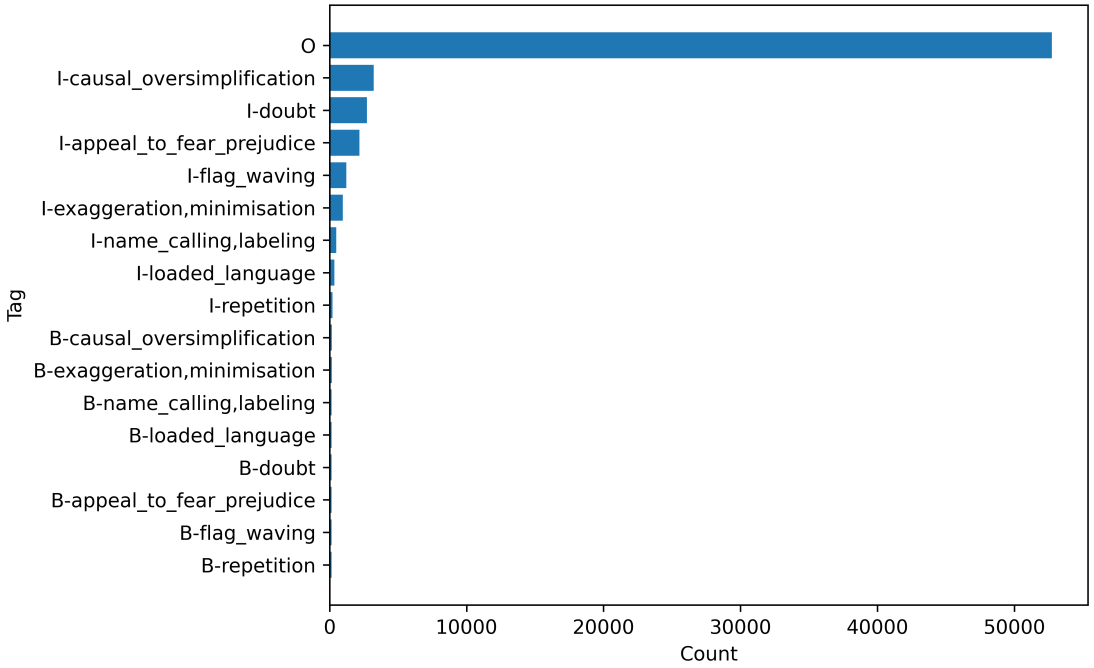


Figure 2. BIO Tag Frequency Distribution

Hyperparameters

Evaluation Metrics

Approach B: BIO Tagging with DistilBERT + CRF Output Layer

Results

Task 1: Propaganda Technique Classification

Multinomial Naive Bayes Accuracy: 0.47

Label	Most Discriminative Features
appeal_to_fear_prejudice	if, we, will, threatened, face, fear, must, be, warned, iran
causal_oversimplification	because, they, she, clearly, it, that, cia, the cia, of needles, needles
doubt	?, why, but, t, guardian, ?, evidence, he, what, story
exaggeration,minimisation	the most, most, history, thing, ever, the world, the worst, world, totally, record
flag_waving	our, american, american people, the american, americans, america, people, country, first, protect
loaded_language	violently, damn, wrath, outright, agenda, violence, head, reckless, rampant, satanic
name_calling,labeling	leftist, infidels, unbelievers, liar, partisan, rapacious, a, gang, disgrace to, priest
repetition	autonomy, aliens, autonomous, annihilated, illegal aliens, indoctrinate, indoctrination, molester, maga, fruitful

Table 3. Most Discriminative Words by Label

BERT Accuracy: 0.53

Task 2: Propaganda Span and Technique Identification

Conclusion

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